Dheeraj Agarwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.13
Vandana Academy Sikshan Samiti	Class XII (RBSE)	2023	94.40%
Nirman Public School	Class X (RBSE)	2021	97.67%

WORK EXPERIENCE

Oracle | Software Engineering Intern

Project Financial Planning Control -- Oracle Fusion AI Studio

- Designed and developed an **AI-powered Agentic App** in Oracle Fusion AI Studio to help project managers monitor financial planning risks across tasks, billing milestones, and change orders.
- Built **workflow orchestration** integrating AI agents, deterministic code, routing, parallel data fetching, and LLMs to generate grounded financial insights and answer project-planning queries.
- Integrated **Oracle Fusion Project Management REST APIs** through a reusable Business Object layer to fetch project, task, financial plan, budget, forecast, billing, invoice, and change-order data.
- Implemented **structured UI widgets and guarded writeback flows** for insight tables, update forms, planning updates, and change-order approval actions, with enterprise AI guardrails.

PROJECTS

Lightweight CounteMeasures for shared LLC Attacks using Cache Partitioning (Team of 2)

- Designed and implemented **UCP-RI** and **UCP-CI**, two lightweighted extensions to **Utility-Cache Based Partioning(UCP)** that invalidate stale cache blocks during repartitioningto mitigate shared LLC side-channel attacks.
- Evaluated across 50 multi-progammed **SPEC CPU2006/2017** and graph workloads in **Champsim** Simulator, achieving up to **3.3% weighted speedup** over **LRU** while preserving cache isolation between co-located processes.

Meeting and Communication Management System (Team of 6)

- Engineered a full-stack web application using React, Websockets, Node.js, PostgreSQL, and the Jitsi API to facilitate real time video conferencing, synchronised agenda tracking, and collaborative task management.
- Integrated a python-based AI intelligence layer utilizing the Sarvam AI SDK and LLMs for bilingual audio transciptions, real time meeting summarization, and automatedaction-item extraction.

RISC-V Simulator

- Developed a **RISC-V simulator in C++** supporting instruction decoding, register updates, memory management, and pipeline simulation.
- Validated simulator behavior by executing test programs and comparing outputs against expected results.

Airline System Management

- Developed an Airline System Management project to handle flight bookings, passenger records, and schedule management.
- Implemented features for ticket reservation, cancellation, and real-time seat availability tracking.
- Designed a user-friendly interface to streamline airline operations and improve customer experience.

Virtual Memory Management in gemOS

- Extended **gemOS** to simulate Linux-style virtual memory using mmap, munmap, **page fault handling**, and demand paging.
- Implemented **Copy-on-Write during fork()**, enabling efficient page sharing between parent and child processes with duplication on write.

SKILLS

Languages : C/C++, python, SQL, Javascript

Libraries/FrameWorks: Numpy,Pandas,Scikit-learn,Pytorch,Tensorflow,Matplotlib

Tools: Git,Docker,VS Code,Bash,LaTeX,npm

Database Skills: SQL, PostgreSQL, MongoDB(NOSql)

Web Development: React, Next.js, HTML, CSS, Javascript

RELEVANT COURSES

- **Core CS:** Software Development Fundamentals, Data Structures & Algorithms, Operating Systems, DBMS, Computer Architecture, Compilers, Software Engineering
- **Theory & Mathematics:** Theory of Computation, Discrete Mathematics, Probability & Statistics, Linear Algebra
- **Artificial Intelligence:** Introduction to AI, Probability and Random Variable, Deep Learning

PROGRAMMING ACHIEVEMENTS

- **Codeforces:** Maximum Rating **1316** | Best Rank **2373** in Round 1036.
- **LeetCode:** Solved **435+ problems** in C++, demonstrating proficiency in **Data Structures, Algorithms, and competitive problem solving**.

EXTRACURRICULAR

Sports & Games: Enjoy playing volleyball, cricket, indoor games and card games.

Reading: Enjoy reading **Hindi and English novels** across a variety of genres.

POSITIONS OF RESPONSIBILITY

Teaching Assistant, **CS3320(Compilers 1)**, **CS3510(Operating Systems 1)**, and **CS3550(DBMS 1)**, IIT Hyderabad